MACHINE LEARNING AND DEEP LEARNING

Lab 07: Denoising Autoencoders and Variational Autoencoders

Dr. Giacomo Lucato, a.y. 2025-2026

RECALL: SELF-SUPERVISED LEARNING

- **Self-Supervised Learning:** Machine Learning branch consisting of pre-training a model on **pretext tasks,** forcing it to learn useful features representations before being applied to the actual classification task.
- **Example of pre-text tasks:** rotation based (predict the rotation angle of images randomly rotated), jigsaw puzzle (reconstruct an image using a set of tiles), …
- Extremely useful when labeled samples are limited as it learns features before actual classification, allowing for faster and stronger convergence during the mainstream task.

AUTOENCODERS (AE)

- An **Autoencoder** is a deep learning model able to extract features from a given image and use those features to reconstruct the original input.
- It is composed of:
  - An **Encoder:** usually a standard CNN, this module is used to **extract a features representation** from the input image.
  - A **Decoder:** an inverted CNN, it is used to **reconstruct the original input** starting from the features extracted by the Encoder.

- When the model is able to **minimize the difference** between input and reconstructed output, we can **discard the Decoder** and keep the Encoder as pre-trained feature extractor.

TRANSPOSED CONVOLUTION (DECONVOLUTION)

- In standard CNNs, the input image is **downsized** during convolutions and max pooling operations.
- When the Decoder reconstructs the input, it takes the downsized feature representation produced by the Encoder and **recovers the original resolution** by using a special operation called **Transposed Convolution.**
- **Transposed Convolution** takes the value of a pixel and multiplies it by a **learnable kernel,** spreading out the value of the pixel, recovering lost resolution.

DENOISING AUTOENCODERS

- **Denoising Autoencoders** are a more stable version of standard Autoencoders.
- The architecture is the same, but instead of starting from a clean image coming from the dataset, they use a modified version, with added **Gaussian noise.**
- The model extracts features from the noisy version of the image, uses them to reconstruct the input, and the **loss** is computed between the model's output and the **original, clean image.**
- The model is therefore forced to extract features **independent from the noise:** this helps the model learn **more stable features**, ignoring random changes in the images.

VARIATIONAL AUTOENCODERS (VAE)

- **Variational Autoencoders** are a version of Autoencoders used not only to learn features by reconstructing the exact same input they received, but also to **generate new data** learning the distribution of features describing the training set.
- These model can be used both as **feature extractor** and as actual **generators of synthetic samples.**
- The internal architecture of these models is composed of usual Encoder and Decoder, but it includes also **two additional learnable parameters** used to represent the features distribution of the dataset: **mean** and **standard deviation** of the distribution.

VARIATIONAL AUTOENCODERS (VAE)

- Variational Autoencoders aim to learn the distribution of features of the training set.
- To generate a new item, they simply **sample randomly** from this learned distribution, creating a new sample never encountered during training.

VARIATIONAL AUTOENCODERS (VAE) - LOSS FUNCTION

- To train a Variational Autoencoder we usually combine two losses together: **Binary Cross Entropy** and **KL Divergence.**
- **BCE** is computed between the original input and the generated output: it measures the **differences** between original pixels and generated pixels, giving a penalty if the differences are excessive.
- If we use BCE alone, the model will simply **memorize training samples,** without learning the ability of generating new ones.
- To avoid this behavior, we add another component to the loss: **KL Divergence** computed between predicted distribution and N(0, 1).
- This forces the model to learn a **smooth, cohesive features distribution**, instead of just memorizing points or sharp peaks corresponding to training samples.

EXERCISE

- The exercise consists of an implementation of a Denoising Autoencoder and a Variational Autoencoder, both trained and evaluated over the MNIST dataset.
- **Task:** download *AE_VAE.ipynb* and follow the comments provided within the notebook.